

SmartDaaS: Methodological Assessment and Field Positioning

1. PROBAST Risk of Bias Self-Assessment

The PROBAST (Prediction model Risk Of Bias Assessment Tool) framework evaluates prediction models across four domains: Participants, Predictors, Outcome, and Analysis. The assessment below maps SmartDaaS against each PROBAST signalling question, with justifications drawn from the published manuscript (under review, Scientific Reports) and supporting repository materials (github.com/Kchinthala15/smartdaas-hiv-validation).

#	Item	Response	Risk of Bias	Justification
1	PARTICIPANTS			
1.1	Were data collected appropriately for the prediction model?	Yes	Low	Data from the Quality of Care HIV dataset (Ogbonna, 2022): de-identified, retrospective, real-world programme records from 27,288 patients enrolled on ART across 11 facility groups in the Nigerian national HIV programme. Data collection predates model development; no prospective contamination.
1.2	Were participants recruited consecutively or randomly?	Yes	Low	The Quality of Care HIV dataset contains all 27,288 HIV-positive patients on ART across the included facilities during the study period (July 2006–December 2018). The dataset is a complete programmatic extract — not a sample or convenience subset — and was collected prior to and independently of this study. No post-hoc selection of participants based on outcome was possible.

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1.3	Were patients excluded appropriately?	Yes	Low	Exclusions limited to patients with insufficient follow-up data to determine outcome status. Exclusion criteria documented in full in supplementary materials.
1.4	Was the study period specified?	Yes	Low	Training period: July 2006–September 2016. Temporal holdout period: October 2016–December 2018. External validation: nine PHIA surveys across six countries (2015–2022).
2	PREDICTORS			
2.1	Were predictors defined and measured prior to the outcome?	Yes	Low	Section 3.3 explicitly states that 'only baseline variables recorded prior to the adherence outcome were included as features.' The outcome (ArvAdherenceLatestLevel = Poor at last visit) is the final event in each patient record. All predictors — including the dominant predictor had_interruption, which reflects historical ART interruption events — were temporally separated from the outcome by a mean of 448 days (confirmed by ablation: AUC difference 1.4% without this feature). No predictor is derived from the same clinical encounter as the outcome.
2.2	Were all predictors available at the time of model use?	Yes	Low	All 15 features are routinely captured in standard HIV programme EMR systems

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				(AMRS, KenyaEMR, DHIS2). No additional data collection is required for deployment.
2.3	Were predictor definitions the same in training and validation?	Yes	Low	Feature definitions consistent across training, holdout, and external PHIA validation sets. Variable mapping documented in utils.py (GitHub).
2.4	Was missing data handled appropriately?	Yes	Low	Missingness documented by feature and facility type (CD4: 17–21% across facility groups). Multiple strategies evaluated. SMOTE applied to training folds only, not to test or validation sets, preventing data leakage.
3	OUTCOME			
3.1	Was the outcome clearly defined?	Yes	Low	Primary outcome: ArvAdherenceLatestLevel = 'Poor' (poor ART adherence at last clinical assessment), documented as a leading indicator of subsequent treatment interruption. Operational framing and clinical pathway documented in model card (GitHub).
3.2	Was the outcome measured without knowledge of predictor values?	Yes	Low	Outcome determined from routine clinical records at a visit subsequent to the predictor measurement period. No prospective outcome ascertainment during model development.

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3.3	Was the time between predictor measurement and outcome appropriate?	Yes	Low	Temporal validation separates training and test periods by calendar time (pre/post September 2016), reflecting realistic deployment conditions. Mean predictor-to-outcome separation: 448 days for the primary predictor.
4	ANALYSIS			
4.1	Were there a sufficient number of participants with the outcome?	Yes	Low	940 outcome events (poor adherence) in 27,288 patients (3.4% prevalence). Events Per Variable = 62.7 (940 events / 15 predictors), substantially exceeding the commonly cited minimum of 10 EPV. Total event count of 940 also exceeds the PROBAST 2019 recommended minimum of 200 events. SMOTE applied to training folds only to address class imbalance; hold-out and temporal validation sets evaluated under the natural class distribution (3.4%).
4.2	Were continuous predictors handled appropriately?	Yes	Low	Continuous predictors (age, CD4, weight, BMI, days-to-ART) retained in continuous form. No arbitrary categorisation. Random Forest handles non-linear relationships and interactions without distributional assumptions.
4.3	Were all candidate predictors handled in the same way?	Yes	Low	Identical preprocessing pipeline applied to all features via utils.py (GitHub). No post-hoc feature selection after observing results.

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4.4	Were selection of predictors based on the full dataset?	Yes	Low	Feature selection performed on training data only within each CV fold. No information leakage from test or holdout sets.
4.5	Was internal validation performed?	Yes	Low	10-fold stratified cross-validation (n=27,288). Stratification preserves class distribution across folds. 95% confidence intervals reported for all performance metrics.
4.6	Were performance measures reported appropriately?	Yes	Low	AUC-ROC, sensitivity, specificity, Brier score, calibration curves, and Decision Curve Analysis (DCA) all reported. DCA confirms net clinical benefit over treat-all and treat-none from thresholds 0.03–0.45.
4.7	Was model calibration assessed?	Yes	Low	Platt scaling and isotonic regression calibration reported. Calibration assessed in both internal holdout and temporal validation sets. This addresses the most common high-risk bias domain across the published literature (Kwarah et al., 2025).
4.8	Was external validation performed?	Yes	Low	Three-level external validation: (1) 20% random holdout (n=10,540); (2) temporal holdout — trained on pre-Sept 2016 data (n=19,084), tested on post-Sept 2016 data (n=8,179); (3) nine independent PHIA population surveys across six countries (n=10,113 adults). Only two of 12 models in the published literature (Kwarah et al., 2025) performed

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				any external validation; none validated across multiple countries.
4.9	Were subgroup analyses performed?	Yes	Low	Subgroup AUC reported across sex, age strata (<30, 30–49, 50+), CD4 count categories (<200, 200–350, >350), and WHO clinical stage. AUC range: 0.815–0.866 (maximum difference 0.051). No systematic disadvantage to any subgroup, including highest-risk CD4 <200 group (AUC 0.829).

Overall PROBAST rating: Predominantly low risk of bias across assessed PROBAST signalling domains. *This contrasts with 75% of models in the published literature (Kwarah et al., 2025, BMC Global and Public Health) showing high risk of bias in at least one domain.*

2. SmartDaaS vs. the Published Literature

The most comprehensive systematic review of machine learning models for HIV treatment interruption prediction to date (Kwarah et al., 2025, BMC Global and Public Health, DOI: 10.1186/s44263-025-00184-4) screened 116,672 records and identified 9 studies reporting 12 models (a further model published in February 2026 — Salami et al., JMIR Medical Informatics — brings the total to 13). The table below maps SmartDaaS against the methodological criteria and gaps identified by that review.

Methodological criterion	Published literature (Kwarah et al., 2025 — 12 models)	SmartDaaS
External validation	2 of 12 models (single country each)	3-level: holdout, temporal, 9 PHIA surveys/6 countries (cross-sectional proxy validation; longitudinal EMR validation planned through APIN/AMPATH pilots)
Mean AUC (internal)	0.668 (SD 0.066) — Kwarah et al., 2025	0.975 (CV); 0.772 temporal holdout — on a stricter test than any reviewed model
Calibration	Absent in 75% of models	Platt scaling and isotonic regression, assessed in both holdout and temporal validation sets
Decision curve analysis	Absent in 75% of models	DCA confirms net benefit across thresholds 0.03–0.45
Missing data handling	High risk of bias in majority	Documented by facility type; SMOTE to training folds only; test/validation sets untouched
Subgroup fairness	Not reported in any reviewed model	AUC 0.815–0.866 across sex, age, CD4, WHO stage (max difference 0.051)
Explainability	Not reported in any reviewed model	Per-patient SHAP attribution for every risk flag
Operational deployment	Prior models include single-country deployments only	Live platform; runs on existing EMR data; no new infrastructure required

Methodological criterion	Published literature (Kwarah et al., 2025 — 12 models)	SmartDaaS
Local recalibration	Not addressed in any reviewed model	Built-in design feature; supported by cross-country variability analysis (9 PHIA surveys)
Sociocultural predictors	Explicitly called for by Kwarah et al.	Cross-country socioeconomic exploratory analysis completed; LGA/community location incorporated into pilot design

Reference: Kwarah W, Vroom FBdC, Dwomoh D, Bosomprah S. Evaluating predictive performance, validity, and applicability of machine learning models for predicting HIV treatment interruption: a systematic review. BMC Global and Public Health. 2025;3:64. doi:10.1186/s44263-025-00184-4

3. Summary for Reviewers and Programme Partners

Every methodological gap identified by the most recent systematic review of this literature — external multi-country validation, calibration, decision curve analysis, explainability, subgroup fairness, missing data transparency, and operational deployment — is addressed by SmartDaaS.

The field's mean internal AUC is 0.668. SmartDaaS achieves 0.772 on temporal holdout validation — a stricter test (trained on earlier data, tested on future data) than the internal cross-validation used by all 12 reviewed models. External validation across nine population-based surveys spanning six countries has not been previously identified in this literature.

To our knowledge, SmartDaaS is among the first identified HIV treatment-retention prediction platforms combining: (1) multi-country external validation (9 PHIA surveys, 6 countries); (2) calibration analysis; (3) decision curve analysis; (4) per-patient SHAP explainability; (5) subgroup fairness assessment; (6) a live deployed platform requiring no new infrastructure; and (7) local recalibration support at each implementing partner. Prior models include single-country validations and single-country deployments, but this full methodological and operational combination has not been previously identified in the published literature. Prospective validation on longitudinal programme EMR data is planned as the primary objective of the APIN and AMPATH pilot studies.

These properties position SmartDaaS not as an incremental contribution to a crowded field, but as the first model in this space to address the full stack of methodological and operational requirements for real-world deployment in PEPFAR-supported HIV programmes.

Platform: smartdaas-hiv-validation.onrender.com

GitHub: github.com/Kchinthala15/smartdaas-hiv-validation

Preprint DOI 1: doi.org/10.64898/2026.05.15.26353325

Preprint DOI 2: doi.org/10.64898/2026.05.15.26353326